



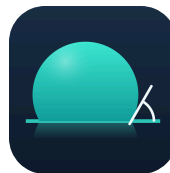
B.TECH PROJECT (MM492) · DEPARTMENT OF METALLURGICAL ENGINEERING & MATERIALS SCIENCE

Automated Sessile-Drop Contact Angle Analyzer

Scientific-precision contact-angle metrology on a smartphone —
designed, built, and quantitatively validated

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IIT Bombay · July 2026



MOTIVATION

Wettability is fundamental; measuring it is expensive

- The contact angle θ is the primary observable of **wetting** — coatings, surface treatments, adhesion, superhydrophobicity.
- Commercial goniometers (Krüss, Biolin, ramé-hart, DataPhysics) reach $\pm 0.1^\circ$ — at $\approx$ **₹15 lakh** per instrument.
- Key insight: they achieve this with modest **1–5 MP cameras**. Their accuracy lives in **optics and software**, not the sensor.
- A smartphone sensor exceeds those cameras — so a careful optical bench + rigorous on-device software should approach lab-grade results.

₹18k

fabricated instrument
(stage + backlight + mount)

>95%

cost reduction vs
commercial goniometer

Goal: laboratory-defensible *static* contact angles — with propagated uncertainty and quality flags, not just a number — fully on-device.

BACKGROUND

The physics of the measurement

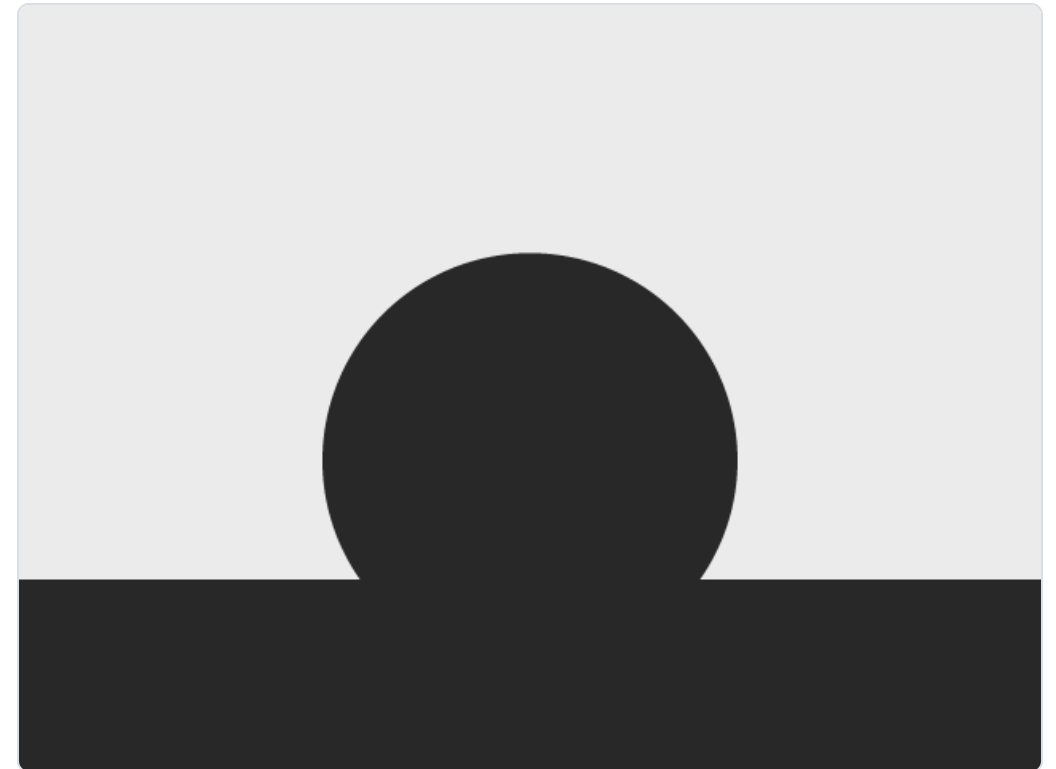
Young's equation — force balance at the three-phase line:

$$\gamma_{SV} - \gamma_{SL} = \gamma_{LV} \cos \theta$$

Young–Laplace (Bashforth–Adams form) — the equilibrium drop profile, arc length s scaled by apex radius b :

$$d\varphi/ds = 2 - \beta z - \sin \varphi / x, \quad \beta = \Delta\rho g b^2/\gamma$$

- $\beta \rightarrow 0$: drop is a **spherical cap**; gravity flattens larger drops.
- Exact geometric bound: $\theta \geq 2 \cdot \text{atan}(h/a)$ — used later as a physical gate on degenerate fits.
- At small β the Young–Laplace fit is **unidentifiable** (angle valid, surface tension not) — flagged, not over-claimed.



The measurement image: a back-lit drop silhouette on its substrate (synthetic 125° cap — angle known exactly, used for validation).

BACKGROUND

Where the error actually comes from

Published error analysis of goniometry (Vuckovac *et al.*, *Soft Matter* 2019):

- **± 1 px of baseline error** $\rightarrow \approx 0.5\text{--}1^\circ$ of angle error below 150° , growing to $\approx 8^\circ$ as $\theta \rightarrow 180^\circ$.
- Error scales **inversely with drop size in pixels**.
- Seven experts analysing the *same image* spread over $\sim 10^\circ$.

Design rules adopted:

- Determine the baseline **algorithmically, to sub-pixel** precision.
- Fill the frame with the drop; warn when it is small.
- **Propagate baseline sensitivity** into every reported uncertainty.

Error budget of optical goniometry

Source	Mechanism	Mitigation here
Baseline placement	dominant; $d\theta/dh = 1/(r \sin\theta)$	4-detector hierarchy, $\$baseline$
Stage / camera tilt	parallax, skewed baseline	live level badge; tilt fit
Highlight clipping	blooming eats the 50% edge	-1 EV bias + capture QC
Autofocus drift	magnification change	tap-to-lock focus
Edge quantisation	± 0.5 px integer edges	sub-pixel 50% crossing
Model bias	circle $\neq$ flattened drop	ensemble + full ADSA

SYSTEM

System overview

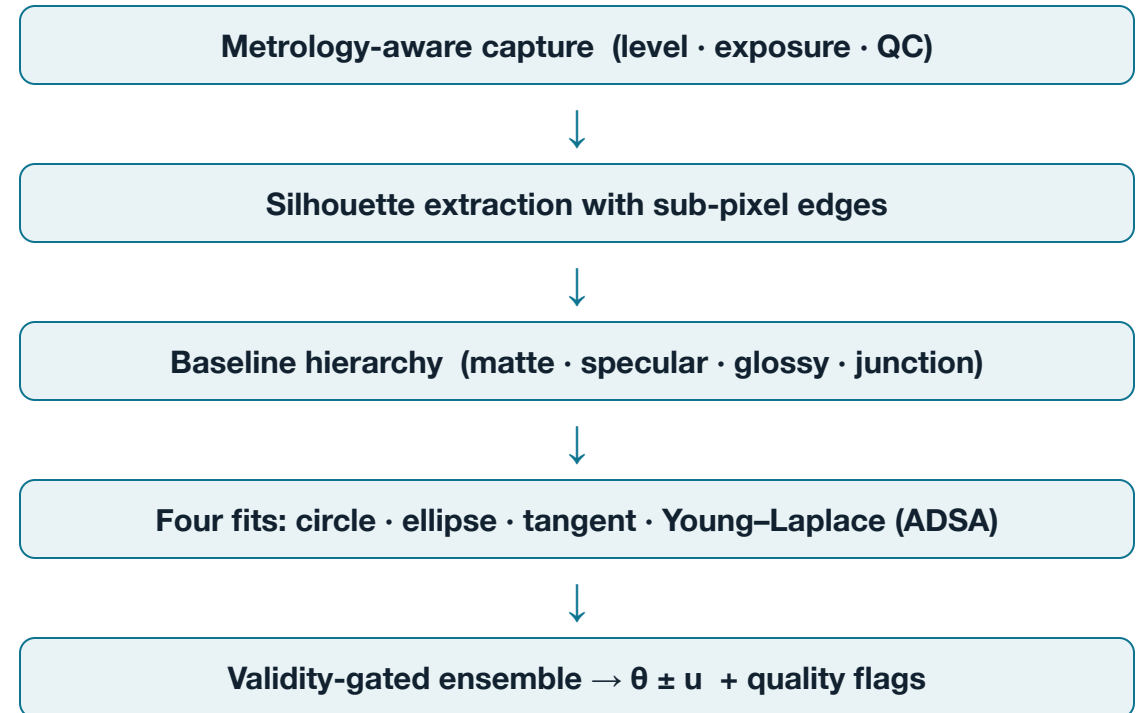
Hardware (fabricated, $\approx$ ₹18,000)

- Uniform LED back-light $\rightarrow$ dark drop on bright field
- X/Z scissor stage for sample positioning
- Multi-axis smartphone mount, acrylic base
- SolidWorks design $\rightarrow$ vendor fabrication

Software

- Flutter/Dart app; analysis backend is **100% native Dart** — no OpenCV in the measurement path, identical on every device and on CI
- 2–5 s per full-resolution image on a mid-range phone
- Annotated overlays, per-method breakdown, CSV export

Analysis pipeline



SYSTEM · HARDWARE

The fabricated instrument



Left: complete integrated setup — LED back-light, sample stage, smartphone mount on an acrylic base. Right: fabricated sample stage — scissor lift (Z) with micrometer translation stage (X).

- **3-block architecture:** uniform LED back-light · adjustable sample holder · multi-axis camera mount.
- Digitally prototyped as a **fully-constrained SolidWorks assembly**.
- **Design-for-fabrication pivot:** sourcing low-volume precision parts locally was prohibitive → custom parametric model, professional vendor fabrication.
- Geometry produces the canonical measurement image: dark silhouette on a bright, even field.
- Total cost $\approx$ ₹18,000 — >95% below a commercial goniometer.

CAPTURE

Metrology-aware capture: discipline before pixels

- **Live attitude badge** from the accelerometer — *roll* (baseline leveling; green $\leq 0.5^\circ$) and *pitch* (optical axis; ISO 19403-2 recommends 0–4° downward so contact points and reflection are visible). Shutter ring turns green only in tolerance.
- **Tap-to-lock focus** — autofocus hunting changes magnification mid-session.
- **–1 EV exposure bias + lock** — bright field kept just below clipping; the sub-pixel edge survives.
- **Zoom from a distance** — poor-man's telecentricity; less perspective error than shooting close.

Post-capture quality control (automatic)

- Highlight clipping > 2% saturated pixels → **blooming** → retake prompt
- Silhouette contrast (p95 – p5) too low → retake prompt

Why it matters: every item maps 1-to-1 onto an entry in the error budget of slide 4 — geometry and exposure faults are cheaper to prevent at capture than to correct in software.

PIPELINE · 1

Silhouette extraction with sub-pixel edges

Why not Canny/Sobel? The dark drop merges into the dark substrate (no edge exists between them); the interior refraction window creates strong *false* edges.

Approach — use the one reliable contrast (bright field vs dark object):

- Otsu threshold → dark objects; connected component **above the baseline** isolates the drop.
- Row-scan of **outermost** pixels → immune to interior artefacts; continues to the per-column baseline so stage tilt cannot amputate the downhill contact.
- Needle-in-drop detected as a sustained narrow apex run and trimmed to the shoulder.

Sub-pixel localisation

For an anti-aliased edge, the true boundary lies exactly at the **50%-coverage intensity crossing** between local background and object levels. Every horizontal edge point *and every baseline point* is refined to this crossing — integer edges cap accuracy at ± 0.5 px, worth $\approx 0.5^\circ$ on small drops since $d\theta \approx \sigma_{\text{baseline}}/a$.

All fits then operate in the **baseline-aligned frame** (true rotation), with left and right angles computed independently and tilt-corrected — the practice of every reference instrument.

PIPELINE · 2

Baseline detection: a hierarchy matched to substrate physics

Detector	Substrate regime	Signal	Origin of the idea
Matte band clustering	dark matte stage (our rig)	per-column first-dark rows cluster at the surface; robust line fit + flank-support test	classical
Specular mirror symmetry	mirror stages (Si, metal)	drop + reflection form a blob symmetric about the surface; sub-pixel symmetry axis + waist test	DropSnake (Stalder 2006)
Glossy step detection	mid-gray glossy (Teflon), camera pitched down	bright→mid-gray plateau-midpoint crossing on object-free flank columns ; first sustained step from top	this work
Junction corner refine	reflective — contact below the visible horizon	corner of each flank's outermost-x trace (V for $\theta < 90^\circ$, waist for $\theta > 90^\circ$); two-segment fits, sub-pixel intersection → baseline and tilt	Krüss ADVANCE · Dropometer · DE10214439A1

Arbitration, validated on real failures: a flank-supported matte line is only overridden by a *corner-confirmed* junction; an unconfirmed horizon loses to any matte line below it; weak fallbacks may never overrule a supported matte line (a circle is locally mirror-symmetric about its equator).

Auditability: every result records *which detector fired* (**baseline_method**) and the fitted tilt.

PIPELINE · 3

Four independent angle-extraction methods

Circle Algebraic LS + geometric R^2 . Exact at $\beta \rightarrow 0$. Gate fixed so hydrophilic caps (centre *below* baseline — correct geometry) are accepted.

Ellipse Halir–Flusser stable direct conic fit; correct eigenvector via full 3×3 characteristic cubic; analytic tangents at the baseline.

Tangent Polynomial fit in a frame **rotated to the local tangent** (weighted PCA): the profile is single-valued at any θ — removes the -5° steep-flank fold bias of image-frame fitting at 145° .

Young–Laplace (ADSA) the physical reference method:

- RK4 integration of Bashforth–Adams from the apex (local error $\approx 10^{-9}$)
- **Orthogonal** residuals; multi-start Nelder–Mead over Bond seeds; robust trimming
- **Contact-zone exclusion:** the optically distorted ≈ 2 px band above the contact is excluded; θ read by **extrapolating** the fitted profile to the baseline (ADSA-NA practice). A proportional band tested *worse* — decisions made by measurement.
- **Degeneracy gates:** reject $\theta < 0.6 \times 2 \operatorname{atan}(h/a)$ (flat-solution artefact); flag $\beta < 0.05$ as circle-degenerate.

A near-perfect ADSA fit **anchors** the ensemble; otherwise methods are combined with quality weights, Tukey outlier moderation and leave-one-out consistency.

PIPELINE · 4

Every angle ships with an uncertainty and quality flags

Five terms, combined in quadrature:

Term	Estimator
Contour sampling	moving-block bootstrap (blocks $\approx \sqrt{n}$ preserve edge-noise correlation)
Method disagreement	$1.4826 \times \text{MAD of valid methods}$
Edge localisation	near-contact residual spread / radius
Contact confidence	calibrated penalty map
Baseline placement	$\sigma_{\text{baseline}} \times d\theta/dh$ (exact, below)

$d\theta/dh = 1/(r \cdot \sin \theta) = 1/a \rightarrow$ reproduces the published $0.5^\circ/\text{px}$ plateau and the $\theta \rightarrow 180^\circ$ blow-up

ISO-style quality flags on the result card

- $\theta \geq 150^\circ$ — baseline sensitivity grows steeply
- $\theta \leq 20^\circ$ — tangent placement ambiguous
- stage tilt $> 2^\circ$ — level the stage
- $|\theta_L - \theta_R| > 4^\circ$ — non-axisymmetric drop or baseline error
- drop < 100 px half-width — error $\propto 1/\text{size}$

The result card also reports **per-method angles, acceptance/rejection reasons, and the baseline detector used** — the measurement is auditable, not a black box.

VALIDATION

Three independent validation tiers — locked as regression tests

Tier 1 — exact synthetic truth. Laboratory references are themselves fits; only synthetics give exact θ . 8×8-supersampled caps, 60–145°, plus noise / tilt / blur / small-drop / shadow stress.

Tier 2 — laboratory reference. 12 PFOTES-coated drops vs LB-ADSA (ImageJ) fits performed in the lab.

Tier 3 — external dataset. Published DropletLab Teflon set: needles, glossy substrate, faint reflections — deliberately adversarial.

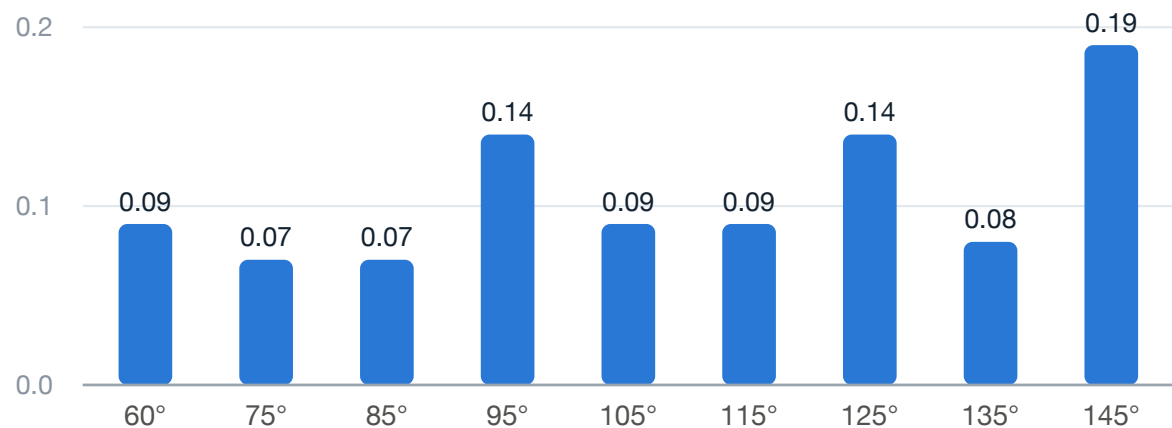
Why this matters:

- Every threshold is **locked into the automated test suite** — any change that degrades accuracy fails the build.
- All improvements this term were made **without regressing** the primary rig: Tables on the next slides are bit-for-bit stable under the Tier-3 work.
- Ground truth was *expanded* (16 transcribed reference labels) before tuning — statistics before optimisation.

VALIDATION · TIER 1

Synthetic truth: MAE 0.106° clean, 0.135° under stress

|ensemble error| (°) vs true angle — clean synthetics

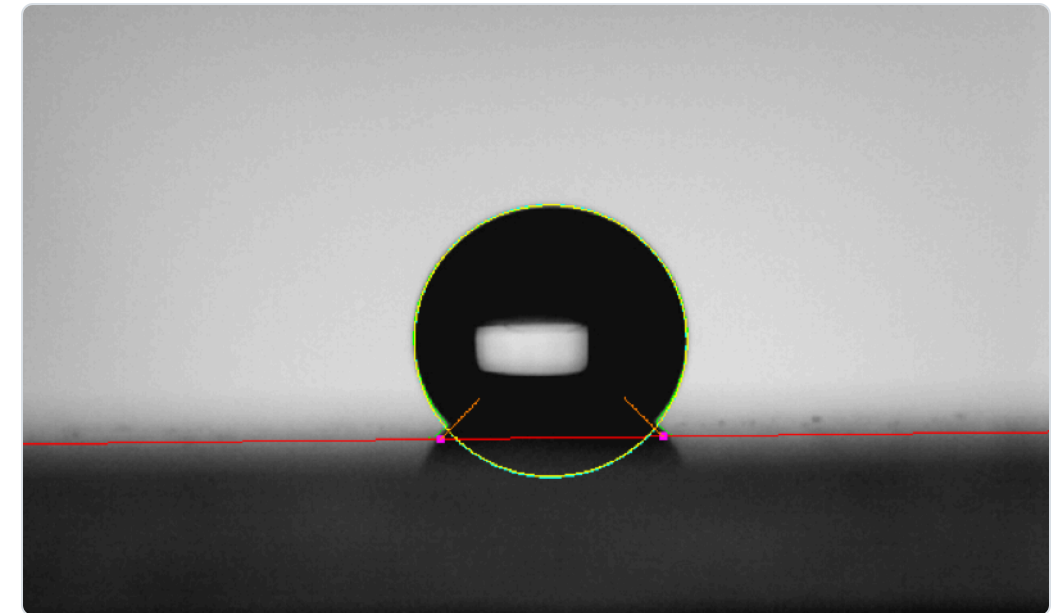
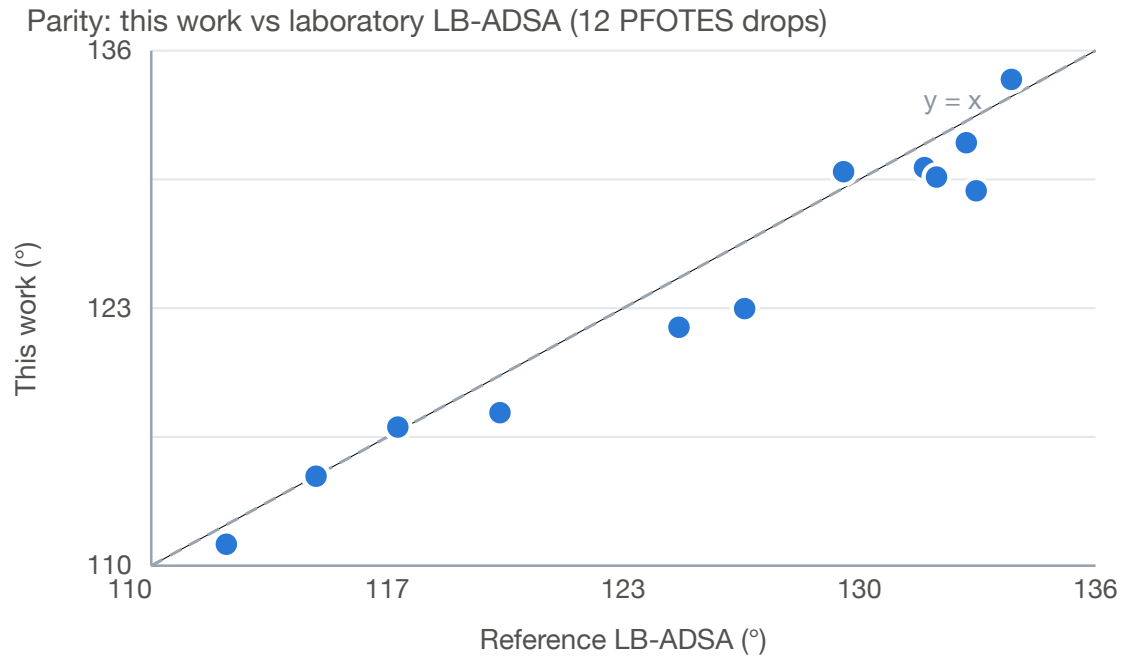


Stress case	θ true	err
Noise $\sigma=8$	125°	0.12°
Stage tilt +5°	120°	0.08°
Small drop R=65 px	140°	0.32°
Defocus blur r=3	140°	0.21°
Contact shadow	150°	0.31°
Contact shadow	135°	0.03°

12 stress cases, MAE **0.135°**, max 0.32°. The +5° tilt case cost **1.76°** before the tilted-baseline contour completion — now 0.08°.

VALIDATION · TIER 2

Laboratory reference: 1.56° MAE against LB-ADSA



Annotated output: contour (green), baseline (red), contacts (magenta), fitted models.

- MAE **1.56°** ($\approx 1.2\%$ relative), max 3.78°; ADSA converged 12/12
- Residuals are **zero-mean scatter** at the reference method's own operator variability — further tuning on 12 samples would over-fit

VALIDATION · TIER 3

Generalisation: 21.2° → 7.3° on an adversarial external dataset

DropletLab Teflon set: dispensing needles, mid-gray glossy substrate, downward camera (visible horizon = substrate's **far edge**, not the contact line), faint reflections, burned-in annotations. 16 transcribed reference angles.

Labelled MAE (°), before → after the baseline research

Before  21.2°

After  7.3°

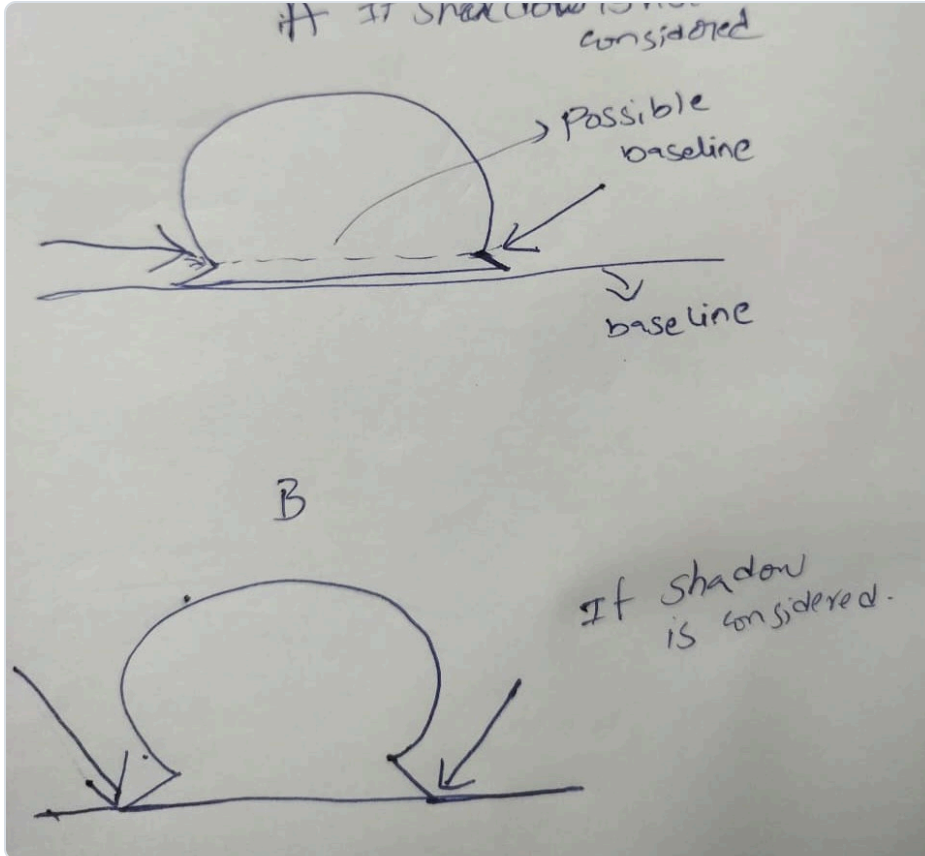
Fixes: glossy step detector · junction-corner refinement · component-selector correction · ADSA degeneracy gate — **zero change** to Tiers 1–2.

Liquid	n	MAE	Max
DI water	7	7.4°	18.6°
Ethylene glycol	4	2.1°	5.6°
Glycerol	5	11.4°	20.7°
Overall	16	7.3°	—

References are themselves polynomial fits on needle-distorted drops (their own L–R asymmetry up to 16°) — 7.3° bounds the **combined** disagreement of both methods.

CASE STUDY • THE PROBLEM

A hazard we identified on our own rig: where is the baseline?



Our working sketch: Case A — shadow not considered, a “possible baseline” appears above the true one; Case B — shadow considered, drop + shadow form wedges at the contacts.

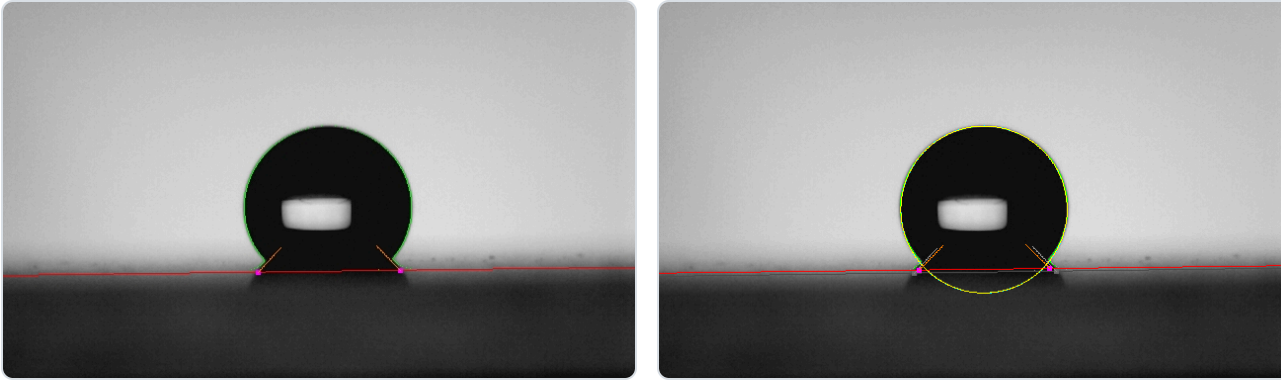
The mechanism. Under the drop's overhang the back-light is blocked: the drop, its shadow and its reflection merge into *one* dark region. Two candidate baselines appear — the true contact plane, and the shadow/reflection boundary below it. At $\approx 0.5\text{--}1^\circ$ per pixel (slide 4), picking the wrong one costs tens of degrees.

Methods we applied to solve it:

- **Specular mirror-symmetry:** treat the reflection as *signal* — the baseline is the sub-pixel symmetry axis of the drop + reflection blob (waist test, $\theta > 90^\circ$).
- **Glossy step detection:** plateau-midpoint crossing on object-free flank columns finds the surface even when it is brighter than the threshold.
- **Junction-corner refinement:** the corner of each flank's outermost-edge trace (slope-sign flip) marks the contact on both sides → baseline *and* tilt.
- **Arbitration + proof:** strict precedence rules between detectors, then synthetic wedge images with exactly known θ locked in as regression tests.

CASE STUDY · THE SOLUTION

The contact-shadow hazard — identified, solved, proven



Left: rig capture — a dark wedge forms under the drop's overhang between the contacts. Right: pipeline result — baseline (red) on the substrate horizon, contacts (magenta) at the wedge apexes.

- Naive detectors place the baseline at the **shadow bottom** → inflated contact width, corrupted θ .
- This pipeline: $\theta = 134.95^\circ$, left-right agreement **0.01°**; circle 135.6° / ADSA 135.9° / ellipse 132.4° — and self-consistent with $a/R = \sin \theta$.
- **Proof, not anecdote:** synthetic wedge + penumbra cases recover 150° to **0.31°** and 135° to **0.03°** — locked as regression tests.

HONEST BOUNDARIES

Known limitations — characterised, not hidden

- **Shadow-skirt regime** (external glycerol images): a soft halo widens *monotonically* into the substrate edge — no wedge apex, waist or corner exists for any trace-geometry detector. Costs 12–20° there. Fix path known: mirrored-model image-energy fit (Stalder) or learned segmentation.
- **Small drops:** error $\propto 1/\text{pixel-size}$ — flagged below 100 px half-width; the remedy is optical (zoom / move closer).
- **$\theta \rightarrow 180^\circ$:** baseline sensitivity $1/(r \sin\theta)$ diverges; results $\geq 150^\circ$ carry a warning, consistent with published advice to prefer force-based methods there.

Future work

- Mirrored-model baseline for the skirt regime
- Needle-side independent fitting (Albert 2019, $\leq 0.6^\circ$)
- Video capture: per-frame angles over a stability plateau (mean $\pm$ SD)
- In-frame scale calibration $\rightarrow$ volume, physical Bond number, surface tension in the identifiable regime
- Small on-device segmentation model as a fallback for messy field images

CONCLUSIONS

A validated instrument, not just an app

0.106°

MAE vs exact synthetic truth (60–145°); 0.135° under stress

1.56°

MAE vs laboratory LB-ADSA — at the reference's own scatter

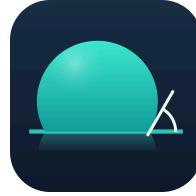
7.3°

on an adversarial external dataset (from 21.2°), zero rig regression

₹18k

instrument cost — >95% below commercial

- Reference physics on-device: full Bashforth–Adams ADSA with orthogonal residuals, exclusion zone, degeneracy gates.
- Baseline detection matched to substrate physics — matte, specular, glossy, reflective — with auditable arbitration.
- Metrologically honest output: propagated uncertainty (baseline term dominant, as the literature requires) + ISO-style flags.
- Every accuracy claim is **reproducible and regression-guarded** in an automated test suite (27 tests).



Thank you

Questions welcome — every number in this talk is reproducible from the project's automated validation suite.

Key references: Young, Phil. Trans. R. Soc. 95 (1805) · Bashforth & Adams (1883) · Rotenberg, Boruvka & Neumann, JCIS 93 (1983) · Hoorfar & Neumann, Adv. Colloid Interface Sci. 121 (2006) · Stalder et al., Colloids Surf. A 286 (2006) & 364 (2010) · Yang, Yu & Zuo, Langmuir 33 (2017) · Vuckovac et al., Soft Matter 15 (2019) · Huhtamäki et al., Nat. Protoc. 13 (2018) · Nurmi et al., Soft Matter 21 (2025) · Chen, Muros-Cobos & Amirfazli, Rev. Sci. Instrum. 89 (2018) · Albert et al., ACS Omega 4 (2019) · ISO 19403-2:2024 · Otsu, IEEE TSMC 9 (1979) · Halíř & Flusser, WSCG (1998) · Nelder & Mead, Comput. J. 7 (1965) · Berry et al., JCIS 454 (2015) · DropletLab dataset (CC-BY)

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